

UNIT	Quantitative Portfolio Management Syllabus for Term Test 1
1	Overview of Portfolio Management: Concept and objectives of Portfolio Management, Types of portfolio management (brief), Role of Portfolio Managers, SEBI regulations related to portfolio operations. Expected Risk and Returns: Portfolio Analysis- Meaning and its Components, Calculation of Expected Return and Risk, Calculation of Covariance, Risk – Return Trade off.
2	Modern Portfolio Theory & Asset Pricing Models: Markowitz Model & Efficient Frontier, Optimal risky portfolio, Capital Market Theory: CML & SML, CAPM (assumptions, applications), Single Index Model, Arbitrage Pricing Theory (conceptual + application), Implementation of efficient frontier, GMV, maximum Sharpe portfolio
3	Modelling Asset Returns: Random Walks, Geometric Brownian Motion(GBM) Simulation Approaches: Monte Carlo simulation of asset prices, generating price paths, Simulating portfolio returns, Scenario analysis